

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 2

Year 12 Higher School Certificate
Trial Examination Term 3 2018

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- A reference sheet is provided separately
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination room

Total marks – 100

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided

Section II Pages 7 – 18

90 marks

Attempt Questions 11 – 16

Start each question in a new writing booklet

Write your student number on every writing booklet

Question	1-10	11	12	13	14	15	16	Total
Total	/10	/15	/15	/15	/15	/15	/15	/100

This assessment task constitutes 45% of the Higher School Certificate Course School Assessment

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

1 The ellipse with equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$ has foci:

(A) $(0, \pm\sqrt{5})$

(B) $(\pm\sqrt{5}, 0)$

(C) $\left(0, \frac{\pm\sqrt{5}}{3}\right)$

(D) $\left(\pm\frac{5}{\sqrt{3}}, 0\right)$

2 The equation $x^3 + 2x^2 + 3 = 0$ has roots α , β and γ .
Which equation has roots 2α , 2β and 2γ ?

(A) $(x-2)^3 + 2(x-2)^2 + 3 = 0$

(B) $2x^3 + 4x^2 + 6 = 0$

(C) $8x^3 + 8x^2 + 3 = 0$

(D) $x^3 + 4x^2 + 24 = 0$

3 The derivative of the implicitly defined curve $x^2 - 2xy + 4y^2 = 12$ is given by $\frac{dy}{dx} = \frac{x-y}{x-4y}$.

The x -values of the points on the curve where the tangents are vertical are:

(A) $x = \pm 1$

(B) $x = \pm 2$

(C) $x = \pm 4$

(D) $x = 1$

- 4 A hyperbola has asymptotes $y = \pm x$ and passes through the point $(3, 2)$.

The Cartesian equation of the hyperbola is:

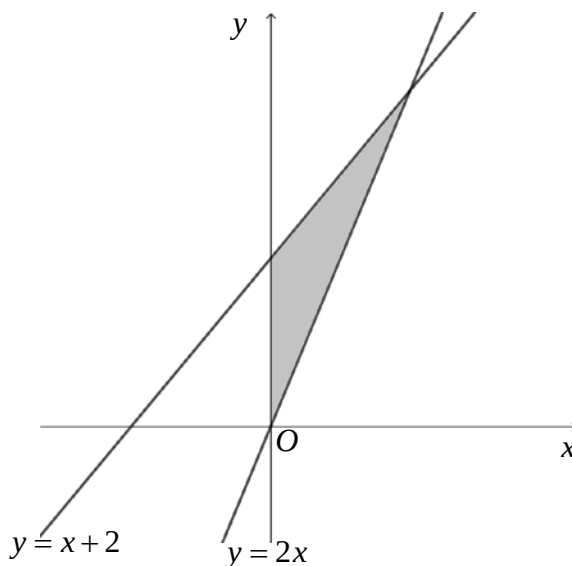
(A) $\frac{y^2}{3} - \frac{x^2}{3} = 1$

(B) $\frac{x^2}{3} - \frac{y^2}{3} = 1$

(C) $\frac{y^2}{5} - \frac{x^2}{5} = 1$

(D) $\frac{x^2}{5} - \frac{y^2}{5} = 1$

- 5 In the diagram below, the region bounded by the lines $y = x + 2$, $y = 2x$ and the y -axis is rotated about the line $y = 5$.



NOT TO
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The volume of the solid of revolution is given by:

(A) $\pi \int_0^2 (16 - 14x + 3x^2) dx$

(B) $\pi \int_0^2 (14x - 16 - 3x^2) dx$

(C) $\pi \int_0^2 (5 - 2x)^2 dx$

(D) $\pi \int_0^2 (3 - 2x)^2 dx$

6 By taking logarithms of both sides, or otherwise, the derivative of $y = x^{\sin x}$ with respect to x is:

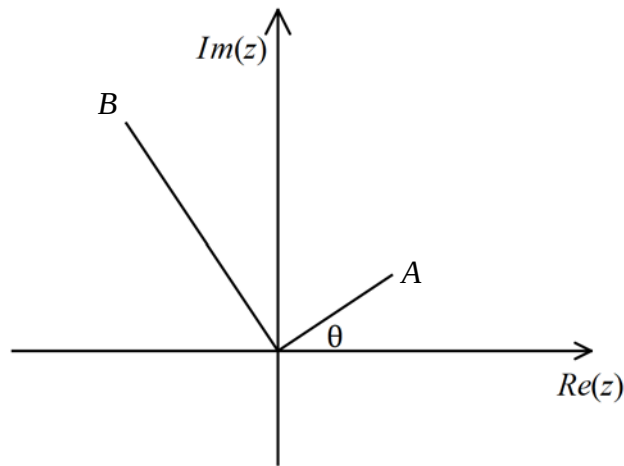
(A) $x^{\sin x - 1} \sin x$

(B) $x^{\sin x} \left(\cos x \log_e x + \frac{\sin x}{x} \right)$

(C) $x^{\sin x} (\cos x \log_e x)$

(D) $\cos x \log_e x + \frac{\sin x}{x}$

7 The points A and B in the diagram represent the complex numbers z_1 and z_2 respectively, where $|z_1| = 1$ and $\arg(z_1) = \theta$ and $z_2 = \sqrt{3}iz_1$.



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$z_2 - z_1$ in modulus-argument form is:

(A) $2[\cos(120^\circ + \theta) + i \sin(120^\circ + \theta)]$

(B) $3[\cos(120^\circ + \theta) + i \sin(120^\circ + \theta)]$

(C) $2[\cos(120^\circ - \theta) + i \sin(120^\circ - \theta)]$

(D) $3[\cos(120^\circ - \theta) + i \sin(120^\circ - \theta)]$

- 8** The equation $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a root of multiplicity 3.
The root of multiplicity 3 is:
- (A) $x = -2$
- (B) $x = \frac{3}{2}$
- (C) $x = -\frac{1}{4}$
- (D) $x = -\frac{1}{2}$
- 9** The value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{\sin x \cos x}$ is:
- (A) $\log_e \sqrt{2}$
- (B) $\log_e \sqrt{3}$
- (C) $\log_e \frac{1}{\sqrt{2}}$
- (D) $\log_e \frac{1}{\sqrt{3}}$
- 10** The minimum value of $f(x) = x \cos^{-1}\left(\frac{x}{3}\right)$ is:
- (A) 0
- (B) -3
- (C) $-\frac{3\pi}{2}$
- (D) -3π

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

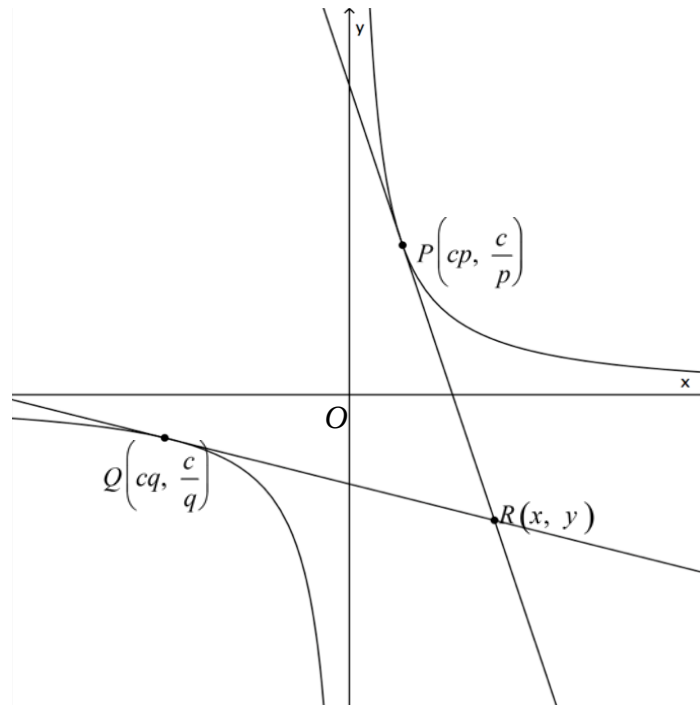
Question 11 (15 marks) Start a new writing booklet

- (a) Let $z = \frac{3-i}{1-2i}$.
- (i) Express z in the form $a+ib$ where a and b are real. 1
- (ii) Hence, express z^4 in the form $a+ib$ where a and b are real. 2
- (b) On the Argand diagram, shade in the region containing all points representing the complex number z such that $|z - (1+i)| \leq 1$ and $\operatorname{Re}(z) > 1$. 3
- (c) Find the square roots of $-5-12i$ in the form $a+ib$ where a and b are real. 2
- (d) If the roots of the equation $x^3 + 4x - 2 = 0$ are α , β and γ , find the value of:
- (i) $\alpha^2 + \beta^2 + \gamma^2$. 2
- (ii) $\alpha^4 + \beta^4 + \gamma^4$. 2
- (e) Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{3+5\cos x}$. 3

Question 12 (15 marks) Start a new writing booklet

- (a) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the rectangular hyperbola $xy = c^2$, $c > 0$.

Tangents drawn at P and Q intersect at $R(x, y)$.



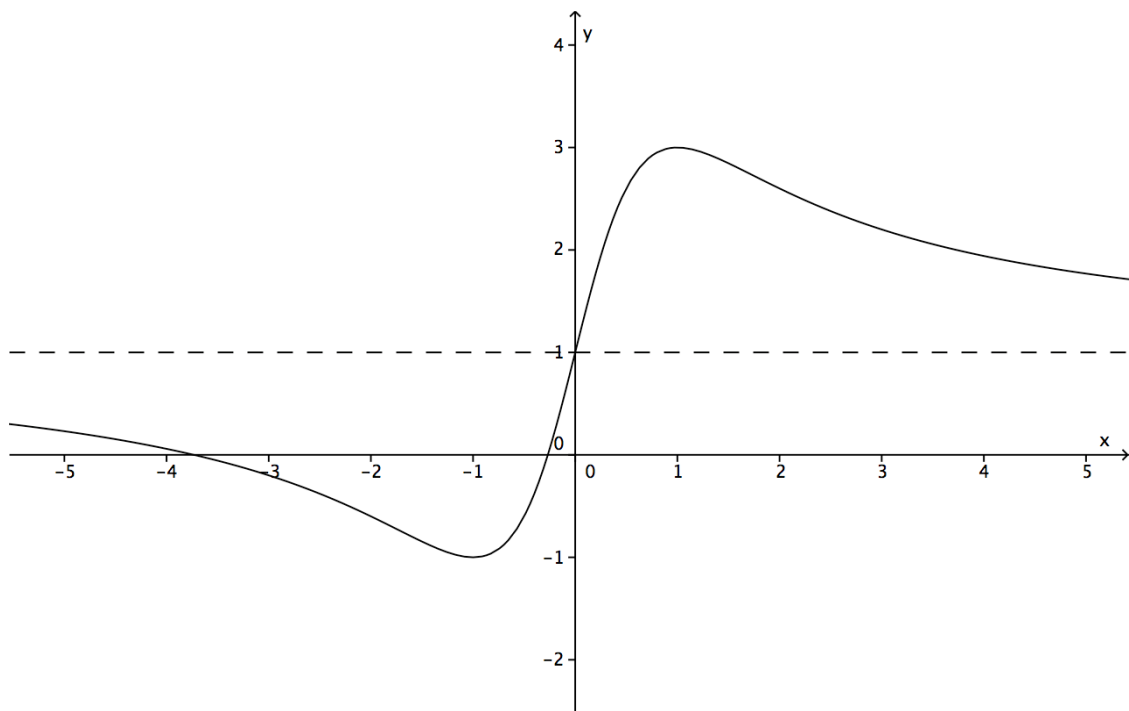
NOT TO
SCALE

- (i) Show that the tangent at P has equation $x + p^2y = 2cp$. 2
- (ii) Show that $R(x, y)$ has coordinates $x = \frac{2cpq}{p+q}$, $y = \frac{2c}{p+q}$. 2
- (iii) If P and Q move on the hyperbola such that $p^2 + q^2 = 2$, show that the Cartesian equation of the locus of R is $xy + y^2 = 2c^2$. 2
- (b) (i) Find A and B such that $\cos x \equiv A(\cos x - 2\sin x) + B(\sin x + 2\cos x)$. 1
- (ii) Hence, find $\int \frac{\cos x}{\sin x + 2\cos x} dx$. 2

Question 12 continues on page 9

Question 12 (continued)

(c) The diagram shows the graph of $y = f(x)$.



Draw a separate half-page diagram for each of the following functions, showing all asymptotes and intercepts.

(i) $y = \sqrt{f(x)}$ 2

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $y = \cos^{-1}[f(x)]$ 2

End of Question 12

Question 13 (15 marks) Start a new writing booklet

(a) By using a substitution, show that $\int_0^1 \frac{x}{(3x+1)^2} dx = \frac{2}{9} \log_e 2 - \frac{1}{12}.$ **3**

(b) (i) Show that $\left(z - \frac{1}{z}\right)^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1}).$ **2**

(ii) By letting $z = \cos \theta + i \sin \theta$, or otherwise, show that **2**
 $32 \sin^5 \theta = 2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta.$

(iii) Hence, find the exact value of $\int_0^{\frac{\pi}{2}} \sin^5 \theta d\theta$ **2**

(c) The polynomial $P(x) = 3x^3 + 4x^2 + 5x - 6$ has only one real zero α , where $|\alpha| < 1$. **2**
Factorise $P(x)$ over the complex field.

(d) A car of mass 800 kg travels along a straight horizontal road. The engine of the car produces a constant driving force of magnitude 2000N. At time t seconds, the speed of the car is $v \text{ ms}^{-1}$. As the car moves, the total resistance to the motion of the car is of magnitude $(400 + 4v^2) \text{ N}$. The car starts from rest.

(i) Show that the acceleration $a = 2 - \frac{v^2}{200}.$ **1**

(ii) Show that $v = 20 \left(\frac{e^{\frac{t}{5}} - 1}{e^{\frac{t}{5}} + 1} \right).$ **2**

(iii) Find the limiting speed of the car, giving reasons for your answer. **1**

End of Question 13

Question 14 (15 marks) Start a new writing booklet

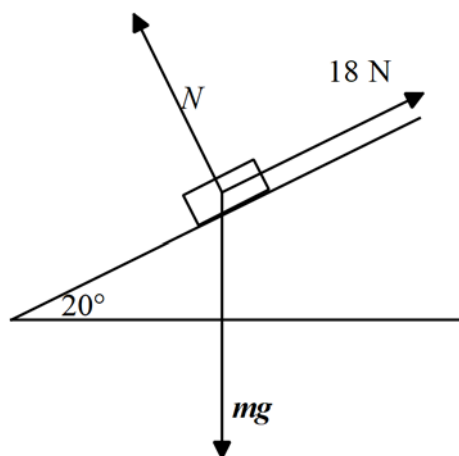
(a) Find $\int \frac{1}{e^x + 4e^{-x}} dx$. 2

(b) It is given that $I_n = \int_0^4 x^n \sqrt{4-x} dx$, where $n \geq 0$, and $\int_0^4 \sqrt{4-x} dx = \frac{16}{3}$.

(i) Show that $I_n = \frac{8n}{2n+3} I_{n-1}$ for $n \geq 1$. 2

(ii) Hence find I_2 . 1

- (c) A box of mass 2 kg is pulled up a rough plane face by means of a light rope. The plane is inclined at an angle of 20° to the horizontal, as shown in the diagram below. The rope is parallel to the slope of the plane. The tension in the rope is 18N.
Use $g = 9.8 \text{ ms}^{-2}$.



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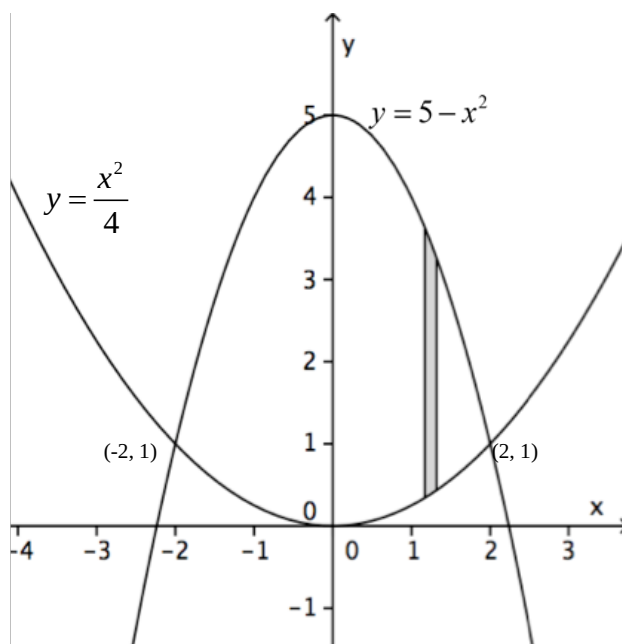
- (i) Show that the normal reaction of the plane on the box is 18.4 N, correct to one decimal place. 1

- (ii) Given that the friction between the box and the plane is $\frac{3}{5}$ of the normal reaction, 2
find the acceleration of the box.

Question 14 continues on page 12

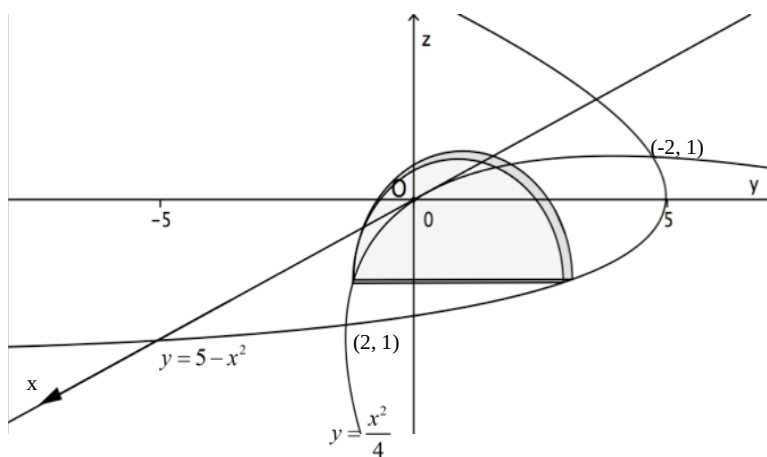
Question 14 (continued)

- (d) The base of a solid is the region bounded by the curves $y = 5 - x^2$ and $y = \frac{x^2}{4}$, as shown in the diagram below. The two parabolas intersect at points $(2, 1)$ and $(-2, 1)$.



NOT TO
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Cross sections by planes perpendicular to the x -axis are semi-circles with the diameter in the base, as shown in the diagram below.



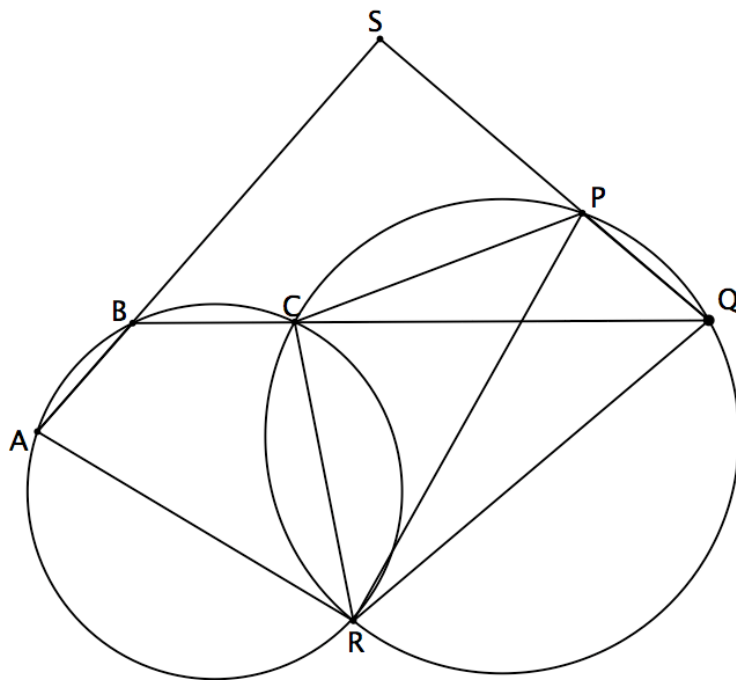
NOT TO
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- (i) Show that the area of the semi-circle is the expression $A = \frac{25\pi}{128}(4 - x^2)^2$. 2
- (ii) Hence, find the volume of the solid. 2

Question 14 continues on page 13

Question 14 (continued)

- (e) In the diagram below, circles $ABCR$ and $CPQR$ intersect at C and R , and BCQ is a straight line. AB produced and QP produced meet at an external point S .



NOT TO
SCALE

Copy or trace the diagram into your writing booklet.

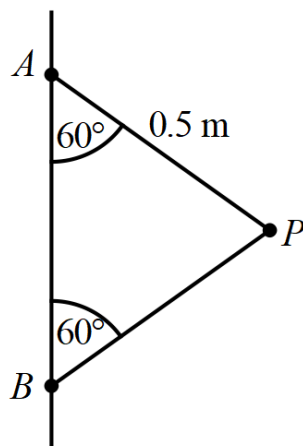
- | | | |
|------|--|---|
| (i) | Prove that $\angle BAR = \angle QPR$. | 2 |
| (ii) | Prove that $ARPS$ is a cyclic quadrilateral. | 1 |

End of Question 14

Question 15 (15 marks) Start a new writing booklet

- (a) The diagram below shows a particle P of mass m attached by two light strings to fixed points A and B , where A is vertically above B . The strings are both taut and P is moving in a horizontal circle with constant angular velocity $2\sqrt{3g}$ radians per second.

Both strings are 0.5 m in length and inclined at 60° to the vertical.



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- (i) If T_A is the tension in the string AP and T_B is the tension in the string BP , show that $T_A - T_B = 2mg$. **1**
- (ii) Find the tensions, T_A and T_B , in the strings in terms of m and g . **2**

Question 15 continues on page 15

Question 15 (continued)

- (b) A particle of mass m is falling vertically under gravity in a resisting medium. The particle is released from rest.

The speed v , in metres per second, of the particle at a distance x from rest is given by

$$v^2 = 2kg \left[1 - e^{\frac{-x}{k}} \right], \text{ where } k \text{ is a positive constant.}$$

- (i) Show that the magnitude of resistance, of the medium is $\frac{mv^2}{2k}$. 2

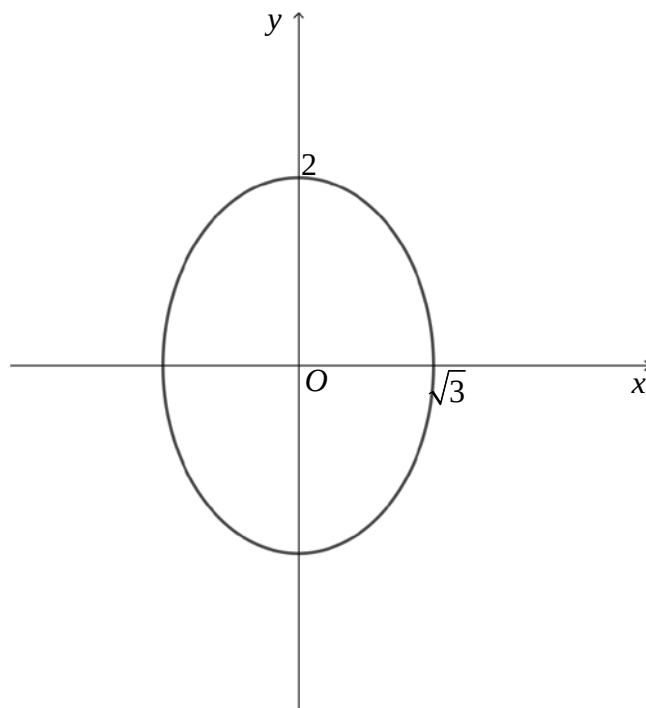
The particle is projected upwards in the same medium with speed $\sqrt{2kg}$.

- (ii) Show that the maximum height reached by the particle above the point of projection is $k \log_e 2$ metres. 2
- (iii) Find the time taken to reach the maximum height above the point of projection. 2

Question 15 continues on page 16

Question 15 (continued)

- (c) The diagram of the ellipse E with equation is $\frac{x^2}{3} + \frac{y^2}{4} = 1$ shown below.



NOT TO
SCALE

The line $y = mx + 4$, where $m > 0$ is a tangent to the ellipse E at the point P .

- (i) Find the value of m . 2
- (ii) Show that the coordinates of P are $\left(-\frac{3}{2}, 1\right)$. 1

The normal at P crosses the y -axis at the point A .

The tangent at P crosses the y -axis at the point B .

- (iii) Find the area of triangle APB . 3

End of Question 15

Question 16 (15 marks) Start a new writing booklet

- (a) One root of the cubic equation $z^3 + az + 10 = 0$, where a is real, is $1 + 2i$.
- (i) Find all three roots of the equation and find the value of a . 2
- (ii) Plot all three roots on the Argand Diagram with a 1:1 horizontal : vertical axes. 1
- (iii) A point z moves in the complex plane such that it lies on the circumference of a circle that passes through the points representing the roots of the equation. 3
Find the equation of the locus of z in the form $|z - z_1| = r$.
- (b) It is known that n is a positive integer, where $n \geq 2$ and that a and b are real and positive.
- (i) Prove that $\frac{a+b}{2} \geq \sqrt{ab}$. 1
- It is known that for positive real numbers $a_1, a_2, \dots$ that $\sqrt[n]{a_1 \times a_2 \times \dots \times a_n} \leq \frac{a_1 + a_2 + \dots + a_n}{n}$.
- (ii) Prove that $n! \leq \left(\frac{n+1}{2}\right)^n$. 2

Question 16 continues on page 18

Question 16 (continued)

(c) It is known that $\sec \theta > \tan \theta$ for $0 \leq \theta < \frac{\pi}{2}$.

(i) Sketch $y = \sec \theta$ and $y = \tan \theta$ for $0 \leq \theta < \frac{\pi}{2}$ on the same number plane. **1**

(ii) Prove the identity $\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$. **1**

(iii) Deduce from (i) and (ii) that $0 < \sec \theta - \tan \theta \leq 1$ for $0 \leq \theta < \frac{\pi}{2}$, giving clear reasons. **2**

(iv) Find the general solution to the equation $\sec \theta - \tan \theta = \frac{1}{2}$. **2**

End of Examination

Year 12 Mathematics Extension 2 Trial Examination Solutions 2018
Multiple Choice

Question 1

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad a = 3, b = 2$$

Eccentricity:

$$b^2 = a^2(1 - e^2)$$

$$4 = 9(1 - e^2)$$

$$e^2 = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3} (e > 0)$$

Foci:

$$S = \left(\pm 3 \times \frac{\sqrt{5}}{3}, 0 \right) \\ = (\pm\sqrt{5}, 0)$$

(B)

Question 2

$$x^3 + 2x^2 + 3 = 0$$

New equation:

$$\left(\frac{x}{2}\right)^3 + 2\left(\frac{x}{2}\right)^2 + 3 = 0$$

$$\frac{x^3}{8} + \frac{2x^2}{4} + 3 = 0$$

$$x^3 + 4x^2 + 24 = 0$$

(D)

Question 3

$$\frac{dy}{dx} = \frac{x-y}{x-4y}$$

For vertical tangents, let

$$x - 4y = 0$$

$$y = \frac{x}{4}$$

Sub into curve:

$$x^2 - 2x\left(\frac{x}{4}\right) + 4\left(\frac{x}{4}\right)^2 = 12$$

$$x^2 - \frac{x^2}{2} + \frac{x^2}{4} = 12$$

$$16x^2 - 8x^2 + 4x^2 = 12 \times 16$$

$$12x^2 = 12 \times 16$$

$$x^2 = 16$$

$$x = \pm 4$$

(C)

Question 4

Note that $y = \pm x$ are asymptotes and hyperbola passes through $(3, 2)$, that is $y < x$.

Therefore hyperbola is sideways.

$$y = \pm \frac{b}{a}x$$

$$b = \pm a$$

$$b^2 = a^2$$

$$\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1$$

Sub $(3, 2)$

$$\frac{9}{a^2} - \frac{4}{a^2} = 1$$

$$a^2 = 5$$

$$a = \sqrt{5} (a > 0)$$

$$\text{Equation is } \frac{x^2}{5} - \frac{y^2}{5} = 1$$

(D)

Question 5

Slices perpendicular to the axis of rotation form annular discs.

$$\begin{aligned} R &= 5 - 2x & r &= 5 - (x + 2) \\ & & &= 3 - x \end{aligned}$$

$$\Delta V = \pi(R^2 - r^2)\Delta x$$

$$= \pi[(5 - 2x)^2 - (3 - x)^2]\Delta x$$

$$= \pi[25 - 20x + 4x^2 - (9 - 6x + x^2)]\Delta x$$

$$= \pi[25 - 20x + 4x^2 - 9 + 6x - x^2]\Delta x$$

$$= \pi(3x^2 - 14x + 16)\Delta x$$

(A)

Question 6

$$y = x^{\sin x}$$

$$\ln y = \sin x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \sin x \times \frac{1}{x}$$

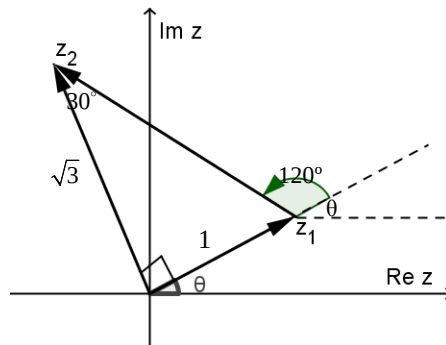
$$\begin{aligned} \frac{dy}{dx} &= y \left(\cos x \ln x + \frac{\sin x}{x} \right) \\ &= x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right) \end{aligned}$$

(B)

Question 7

$$\begin{aligned} \arg(z_2 - z_1) &= 90^\circ + 30^\circ + \theta \\ &= 120^\circ + \theta \end{aligned}$$

$$\begin{aligned} |z_2 - z_1| &= \sqrt{1^2 + \sqrt{3}^2} \\ &= 2 \end{aligned}$$



(A)

Question 8

$$P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$\text{Let } P''(x) = 0$$

$$24x^2 + 54x + 12 = 0$$

$$4x^2 + 9x + 2 = 0$$

$$x = \frac{-9 \pm \sqrt{81 - 32}}{8}$$

$$= \frac{-1}{4} \text{ or } -2$$

$$\begin{aligned} P(-2) &= 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24 \\ &= 0 \end{aligned}$$

(A)

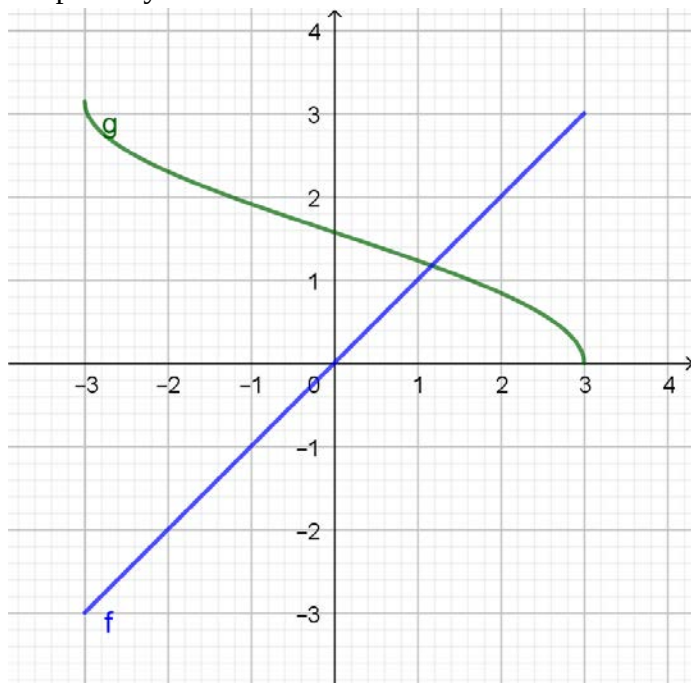
Question 9

$$\begin{aligned}
\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{\sin x \cos x} &= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 x dx}{\sin x \cos x \sec^2 x} \\
&= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan x} dx \\
&= \left[\ln(\tan x) \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\
&= \ln\left(\tan \frac{\pi}{3}\right) - \ln\left(\tan \frac{\pi}{4}\right) \\
&= \ln \sqrt{3} - \ln 1 \\
&= \ln \sqrt{3}
\end{aligned}$$

(B)

Question 10

Graphically:

Minimum will occur at $x = -3$

$$\begin{aligned}
f(-3) &= -3 \times \cos^{-1}(-1) \\
&= -3 \times \pi \\
&= -3\pi
\end{aligned}$$

(D)

Question 11

(a)

(i)

$$\begin{aligned} z &= \frac{3-i}{1-2i} \\ &= \frac{(3-i)(1+2i)}{(1-2i)(1+2i)} \\ &= \frac{3+6i-i-2i^2}{1+4} \\ &= \frac{5+5i}{5} \\ &= 1+i \end{aligned}$$

(ii)

$$\begin{aligned} z &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ z^4 &= \left(\sqrt{2} \right)^4 \left(\cos \pi + i \sin \pi \right) \\ &= 4(-1+0i) \\ &= -4 \end{aligned}$$

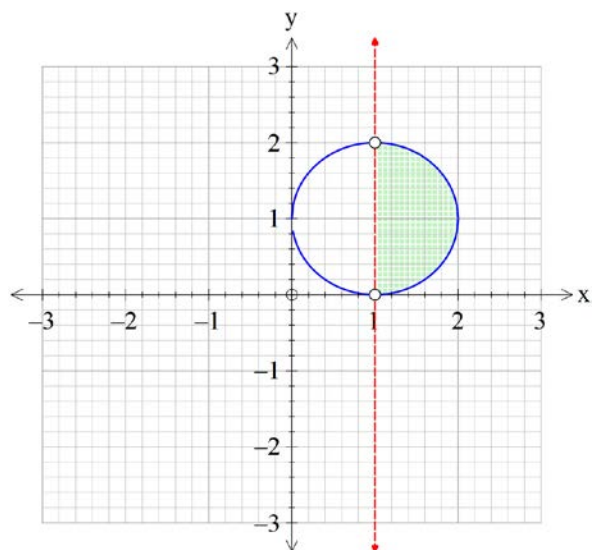
(b)

$$|z - (1+i)| \leq 1$$

Circle centre (1,1) radius 1 unit.

$$\operatorname{Re}(z) > 1$$

$$x > 1$$



(c)

$$\text{Let } (a + ib) = \sqrt{3} + i$$

$$a^2 + 2abi - b^2 = -5 - 12i$$

$$a^2 - b^2 = -5 \dots (1)$$

$$ab = -6 \dots (2)$$

$$a = -\frac{6}{b}$$

$$\frac{36}{b^2} - b^2 = -5$$

$$b^4 - 5b^2 - 36 = 0$$

$$(b^2 - 9)(b^2 + 4) = 0$$

$$b = \pm 3 \text{ (} b \text{ is real)}$$

$$\text{When } b = 3, a = -2$$

$$b = -3, a = 2$$

Therefore the square roots are $2 - 3i$ and $-2 + 3i$.

(d)

$$x^3 + 4x - 2 = 0$$

(i)

$$(\alpha + \beta + \gamma)^2 = (\alpha + \beta)^2 + 2(\alpha + \beta)\gamma + \gamma^2$$

$$= \alpha^2 + 2\alpha\beta + 2\alpha\gamma + 2\gamma\beta + \gamma^2$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= (0) - 2 \times 4$$

$$= -8$$

OR

$$\alpha^3 + 4 - 2 = 0$$

$$\alpha^3 = 2 - 4\alpha$$

$$\alpha^2 = \frac{2}{\alpha} - 4$$

$$\therefore \sum \alpha^2 = 2 \sum \frac{1}{\alpha} - 4(3)$$

$$\therefore \sum \alpha^2 = 2 \frac{\sum \alpha\beta}{\sum \alpha} - 12$$

$$= 2 \left(\frac{4}{2} \right) - 12$$

$$= -8$$

(ii)

$$\alpha^3 + 4 - 2 = 0$$

$$\alpha^3 = 2 - 4\alpha$$

$$\alpha^4 = 2\alpha - 4\alpha^2$$

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = 2(\alpha + \beta + \gamma) - 4(\alpha^2 + \beta^2 + \gamma^2)$$

$$= 2 \times 0 - 4 \times -8$$

$$= 32$$

(e)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{3+5\cos x}, \quad \tan \frac{x}{2} = t$$

$$\frac{x}{2} = \tan^{-1} t$$

$$x = 2 \tan^{-1} t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

$$\text{When } x = \frac{\pi}{2}, \quad t = 1$$

$$x = 0, \quad t = 0$$

$$= \int_0^1 \frac{1}{3 + \frac{5(1-t^2)}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1}{\frac{3(1+t^2) + 5(1-t^2)}{1+t^2}} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{1+t^2}{3(1+t^2) + 5(1-t^2)} \times \frac{2}{1+t^2} dt$$

$$= \int_0^1 \frac{2}{3 + 3t^2 + 5 - 5t^2} dt$$

$$= \int_0^1 \frac{2}{8 - 2t^2} dt$$

$$= \int_0^1 \frac{2}{2(4 - t^2)} dt$$

$$= \int_0^1 \frac{1}{4 - t^2} dt$$

$$= \int_0^1 \frac{1}{(2-t)(2+t)} dt$$

$$= \int_0^1 \frac{1}{2-t} + \frac{1}{2+t} dt$$

$$= \int_0^1 \frac{A}{2-t} + \frac{B}{2+t} dt$$

$$\text{where } \frac{1}{(2-x)(2+x)} \equiv \frac{A}{2-x} + \frac{B}{2+x}$$

$$1 \equiv A(2+x) + B(2-x)$$

$$x = 2, \quad 1 = 4A$$

$$A = \frac{1}{4}$$

$$\text{when } x = -2, \quad 1 = 4B$$

$$B = \frac{1}{4}$$

(e)cont.

$$= \frac{1}{4} \int_0^1 \frac{1}{2-t} + \frac{1}{2+t} dt$$

$$= \frac{1}{4} \int_0^1 -\frac{1}{2-t} + \frac{1}{2+t} dt$$

$$= \frac{1}{4} \left[-\ln|2-t| + \ln|2+t| \right]_0^1$$

$$= \frac{1}{4} \left[\ln \left| \frac{2+t}{2-t} \right| \right]_0^1$$

$$= \frac{1}{4} \left[\ln \left| \frac{3}{1} \right| - \ln \left| \frac{2}{2} \right| \right]$$

$$= \frac{1}{4} \ln 3$$

Question 12

(a) (i) $xy = c^2$ Implicitly differentiating $y + x \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{y}{x}$ At $P\left(cp, \frac{c}{p}\right)$, $\frac{dy}{dx} = -\frac{c}{p} \div cp$ $= -\frac{1}{p^2}$ Equation of tangent: $y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$ $p^2y - cp = -x + cp$ $x + p^2y = 2cp$	
(ii) Equation of tangent at P: $x + p^2y = 2cp \dots (1)$ $x + q^2y = 2cq \dots (2)$ (1)-(2): $y(p^2 - q^2) = 2cp - 2cq$ $y(p - q)(p + q) = 2c(p - q)$ $y = \frac{2c}{p + q}$ Sub into (1) $x + p^2 \times \frac{2c}{p + q} = 2cp$ $x = 2cp - \frac{2cp^2}{p + q}$ $= \frac{2cp^2 + 2cpq - 2cp^2}{p + q}$ $= \frac{2cpq}{p + q}$ $R = \left(\frac{2cpq}{p + q}, \frac{2c}{p + q} \right)$	

(iii)

$$x = \frac{2cpq}{p+q}$$

$$= pq \times \frac{2c}{p+q}$$

$$x = pqy$$

$$\frac{x}{y} = pq \dots (2)$$

$$y = \frac{2c}{p+q}$$

$$p+q = \frac{2c}{y} \dots (4)$$

$$p^2 + q^2 = 2$$

$$(p+q)^2 - 2pq = 2$$

$$\left(\frac{2c}{y}\right)^2 - 2 \times \frac{x}{y} = 2$$

$$\frac{4c^2}{y^2} - \frac{2x}{y} = 2$$

$$4c^2 - 2xy = 2y^2$$

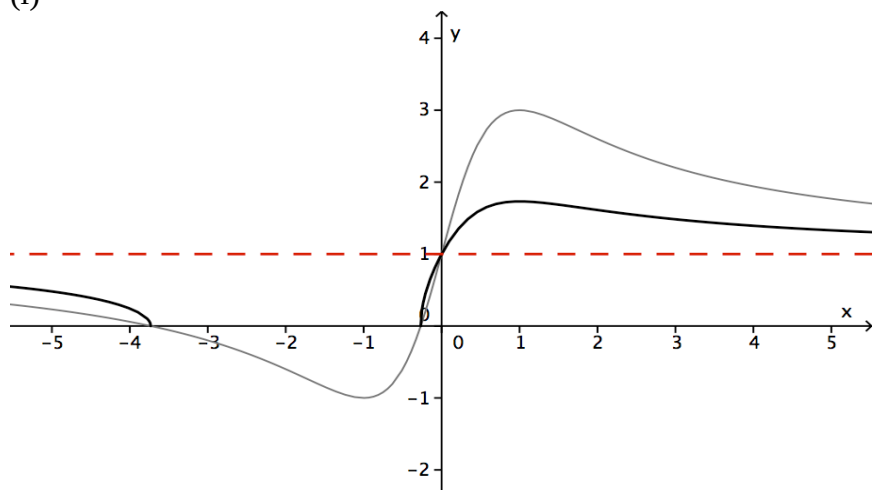
$$2c^2 - xy = y^2$$

$$xy + y^2 = 2c^2$$

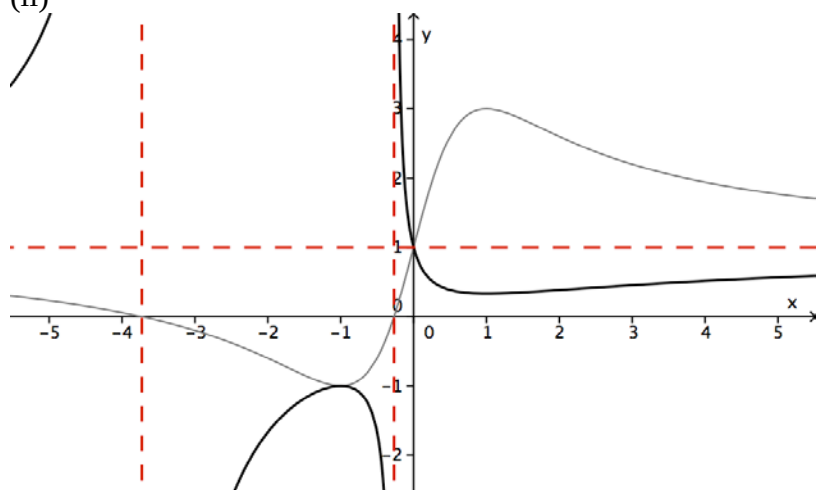
(b)	
(i) $\cos x = A(\cos x - 2 \sin x) + B(\sin x + 2 \cos x)$ $\cos x = A \cos x + 2B \cos x + B \sin x - 2A \sin x$ $\therefore A + 2B = 1$ $B = 2A$ $\therefore A + 4A = 1$ $A = \frac{1}{5}$ $B = \frac{2}{5}$	
(ii) $\cos x = \frac{1}{5}(\cos x - 2 \sin x + 2 \sin x + 4 \cos x)$ $\int \frac{\cos x}{\sin x + 2 \cos x} dx = \frac{1}{5} \int \frac{\cos x - 2 \sin x + 2 \sin x + 4 \cos x}{\sin x + 2 \cos x} dx$ $= \frac{1}{5} \int \left(\frac{\cos x - 2 \sin x}{\sin x + 2 \cos x} + \frac{2(\sin x + 2 \cos x)}{\sin x + 2 \cos x} \right) dx$ $= \frac{1}{5} [\ln(\sin x + 2 \cos x) + 2x] + C$	

(c)

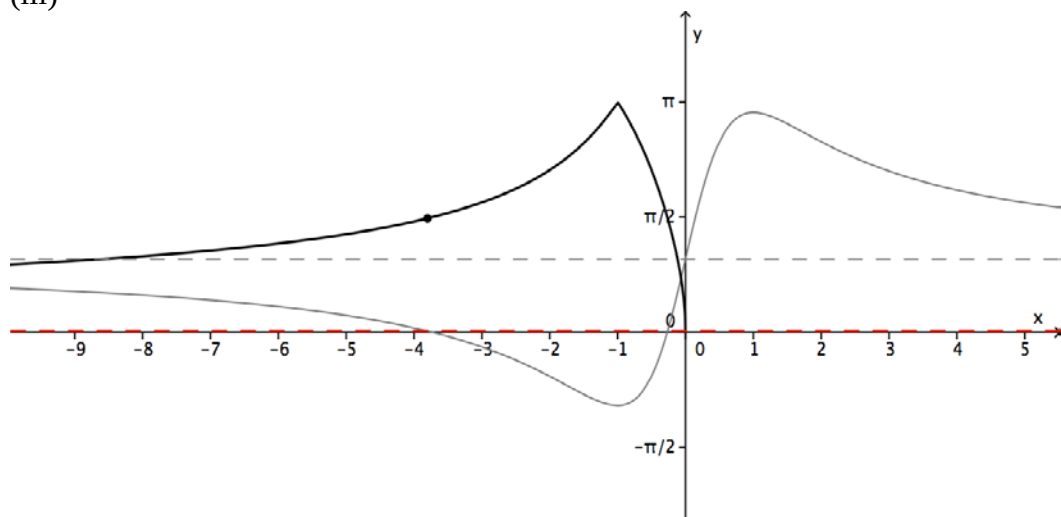
(i)



(ii)



(iii)

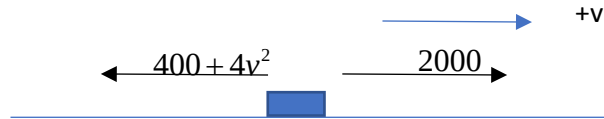


Question 13

(a) $I = \int_0^1 \frac{x}{(3x+1)^2} dx$ Let $u = 3x+1$ When $x=0, u=1$. When $x=1, u=4$. $3x+1=u$ $3x=u-1$ $x = \frac{1}{3}(u-1)$ $u = 3x+1$ $\frac{du}{dx} = 3$ $dx = \frac{du}{3}$ $I = \frac{1}{3} \int_1^4 \frac{(u-1)}{3u^2} du$ $= \frac{1}{9} \int_1^4 \left(\frac{1}{u} - u^{-2} \right) du$ $= \frac{1}{9} \left[\ln u + \frac{1}{u} \right]_1^4$ $= \frac{1}{9} \left[\left(\ln 4 + \frac{1}{4} \right) - (\ln 1 + 1) \right]$ $= \frac{1}{9} \left(\ln 2^2 + \frac{1}{4} - 1 \right)$ $= \frac{1}{9} \left(2 \ln 2 - \frac{3}{4} \right)$ $= \frac{2}{9} \ln 2 - \frac{1}{12}$	
(b) (i) $\left(z - \frac{1}{z} \right)^5 = z^5 - 5 \frac{z^4}{z} + 10 \frac{z^3}{z^2} - 10 \frac{z^2}{z^3} + 5 \frac{z}{z^4} - \frac{1}{z^5}$ $= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$ $= (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$	

(ii) Let $z = \cos \theta + i \sin \theta$ $z^n = \cos n\theta + i \sin n\theta$ $z^{-n} = \cos n\theta - i \sin n\theta$ $z^n - z^{-n} = 2i \sin n\theta$ From (i) $\left(z - \frac{1}{z}\right)^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$ $(z - z^{-1})^5 = 2i \sin 5\theta + 10 \times 2 \sin \theta - 5 \times 2 \sin 3\theta$ $(2i \sin \theta)^5 = 2i \sin 5\theta + 20i \sin \theta - 10i \sin 3\theta$ $2^5 i^5 \sin^5 \theta = i(2 \sin 5\theta + 20 \sin \theta - 10 \sin 3\theta)$ $32i \sin^5 \theta = i(2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta)$ $32 \sin^5 \theta = 2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta$	
(iii) $\int_0^{\frac{\pi}{2}} \sin^5 \theta d\theta = \frac{1}{32} \int_0^{\frac{\pi}{2}} 2(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$ $= \frac{1}{16} \left[-\frac{1}{5} \cos 5\theta + \frac{5}{3} \cos 3\theta - 10 \cos \theta \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{16} \left[\left(-\frac{1}{5} \cos \frac{5\pi}{2} + \frac{5}{3} \cos \frac{3\pi}{2} - 10 \cos \frac{\pi}{2} \right) - \left(-\frac{1}{5} \cos 0 + \frac{5}{3} \cos 0 - 10 \cos 0 \right) \right]$ $= \frac{1}{16} \left[(0 + 0 - 0) - \left(-\frac{1}{5} + \frac{5}{3} - 10 \right) \right]$ $= \frac{8}{15}$	
(c) $P(x) = 3x^3 + 4x^2 + 5x - 6$ Zeros of $P(x)$ are in the form $\pm \frac{6}{3} = \pm \frac{1}{3}, \pm \frac{2}{3}, \quad \alpha < 1$ $P\left(\frac{2}{3}\right) = 3\left(\frac{2}{3}\right)^3 + 4\left(\frac{2}{3}\right)^2 + 5\left(\frac{2}{3}\right) - 6$ $= 0$ Therefore $(3x - 2)$ is a factor of $P(x)$ By short division, $(3x^3 + 4x^2 + 5x - 6) = (3x - 2)(x^2 + 2x + 3)$ $= (3x - 2)(x^2 + 2x + 1 + 2)$ $= (3x - 2)((x + 1)^2 - 2i^2)$ $= (3x - 2)(x + 1 - \sqrt{2}i)(x + 1 + \sqrt{2}i)$ $= (3x - 2)\left(x - (-1 + \sqrt{2}i)\right)\left(x - (-1 - \sqrt{2}i)\right)$	

(d)



(i)

$$F = ma$$

$$2000 - (400 + 4v^2) = 800\ddot{x}$$

$$1600 - 4v^2 = 800\ddot{x}$$

$$\ddot{x} = 2 - \frac{v^2}{200}$$

(ii)

$$\frac{dv}{dt} = \frac{400 - v^2}{200}$$

$$\frac{dt}{dv} = \frac{200}{(400 - v^2)}$$

$$= \frac{5}{(20 - v)} + \frac{5}{20 + v} \quad [\text{by partial fractions}]$$

$$t = 5 \ln(20 + v) - 5(\ln 20 - v) + C$$

$$= 5 \ln\left(\frac{20 + v}{20 - v}\right) + C$$

When $v = 0$,

$$0 = 5 \ln 1 + C$$

$$C = 0$$

$$t = 5 \ln\left(\frac{20 + v}{20 - v}\right)$$

$$\frac{t}{5} = \ln\left(\frac{20 + v}{20 - v}\right)$$

$$e^{\frac{t}{5}} = \frac{20 + v}{20 - v}$$

$$20e^{\frac{t}{5}} - ve^{\frac{t}{5}} = 20 + v$$

$$v\left(1 + e^{\frac{t}{5}}\right) = 20\left(e^{\frac{t}{5}} - 1\right)$$

$$v = 20\left(\frac{e^{\frac{t}{5}} - 1}{e^{\frac{t}{5}} + 1}\right)$$

(iii) As $t \rightarrow \infty$:

$$v = 20 \left(\frac{e^{\frac{t}{5}} + 1 - 1}{e^{\frac{t}{5}} + 1} \right)$$

$$= 20 \left(1 - \frac{1}{e^{\frac{t}{5}} + 1} \right)$$

$$\frac{1}{e^{\frac{t}{5}} + 1} \rightarrow 0$$

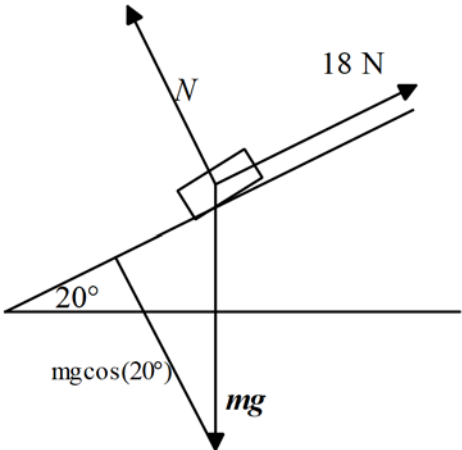
$$v \rightarrow 20$$

The limiting speed is 20 metres per second.

Question 14

(a) $I = \int \frac{1}{e^x + 4e^{-x}} dx$ $= \int \frac{1}{e^x + \frac{4}{e^x}} dx$ $= \int \frac{1}{\frac{e^{2x} + 4}{e^x}} dx$ $= \int \frac{e^x}{e^{2x} + 4} dx$ $= \int \frac{e^x}{(e^x)^2 + (2)^2} dx$ $= \frac{1}{2} \tan^{-1} \frac{e^x}{2} + C$	
(b) (i) $I_n = \int_0^4 x^n \sqrt{4-x} dx$ $u = x^n \quad v' = (4-x)^{\frac{1}{2}}$ $u' = nx^{n-1} \quad v = \frac{-2}{3}(4-x)^{\frac{3}{2}}$ $I_n = \left[\frac{-2}{3} x^n (4-x)^{\frac{3}{2}} \right]_0^4 + \frac{2}{3} n \int_0^4 x^{n-1} (4-x)^{\frac{3}{2}} dx$ $= [0-0] + \frac{2}{3} n \left[\int_0^4 x^{n-1} (4-x) \sqrt{4-x} dx \right]$ $= \frac{2}{3} n \left[4 \int_0^4 x^{n-1} \sqrt{4-x} dx - \int_0^4 x^n \sqrt{4-x} dx \right]$ $= \frac{2}{3} n [4I_{n-1} - I_n]$ $I_n + \frac{2n}{3} I_n = \frac{8nI_{n-1}}{3}$ $I_n \left(1 + \frac{2n}{3} \right) = \frac{8n}{3} I_{n-1}$ $I_n \left(\frac{2n+3}{3} \right) = \frac{8n}{3} I_{n-1}$ $I_n = \frac{8n}{2n+3} I_{n-1}$	

(b)	
(ii) $I_2 = \frac{8}{4+3} I_1$ $= \frac{8}{7} \left(\frac{8}{2+3} I_0 \right)$ $= \frac{8}{7} \times \frac{8}{5} \times \frac{16}{3}$ $= \frac{1024}{105}$	

(c)	
(i)  Resolving forces perpendicular to the plane $N = mg \cos 20^\circ$ $= 2 \times 9.8 \times \cos 20^\circ$ $= 18.417 \dots$ $= 18.4 \text{ N (1dp)}$	
(ii) Resolving forces parallel to the plane: $2a = 18 - \frac{3}{5} \times 18.4 - 2 \times 9.8 \times \sin 20^\circ$ $a = \frac{1}{2} \left(18 - \frac{3}{5} \times 18.4 - 2 \times 9.8 \times \sin 20^\circ \right)$ $= 0.12829 \dots$ $= 0.13 \text{ ms}^{-2}$	

(d)

(i)

$PQ = y_1 - y_2$ and PQ is the diameter of the semi-circle.

$$2r = (5 - x^2) - \frac{x^2}{4}$$

$$2r = 5 - \frac{5x^2}{4}$$

$$2r = \frac{20 - 5x^2}{4}$$

$$2r = \frac{5(4 - x^2)}{4}$$

$$\therefore r = \frac{5(4 - x^2)}{8} \text{ units.}$$

$$\therefore \text{Area of slice: } A = \frac{1}{2} \pi r^2$$

$$= \frac{\pi}{2} \left[\frac{5(4 - x^2)}{8} \right]^2$$

$$= \frac{\pi}{2} \left[\frac{25(4 - x^2)^2}{64} \right]$$

$$\therefore A = \frac{25\pi(4 - x^2)^2}{128} \text{ unit}^2$$

(ii)

Volume of slice:

$$\delta V = \frac{25\pi}{128} (4 - x^2)^2 \delta x$$

$\therefore$ Volume of solid:

$$V = \lim_{\delta x \rightarrow 0} \sum_{-2}^2 \frac{25\pi}{128} (4 - x^2)^2 \delta x$$

$$V = \frac{25\pi}{128} \int_{-2}^2 (4 - x^2)^2 dx$$

$$= \frac{25\pi}{64} \int_0^2 (4 - x^2)^2 dx$$

$$= \frac{25\pi}{64} \int_0^2 (16 - 8x^2 + x^4) dx$$

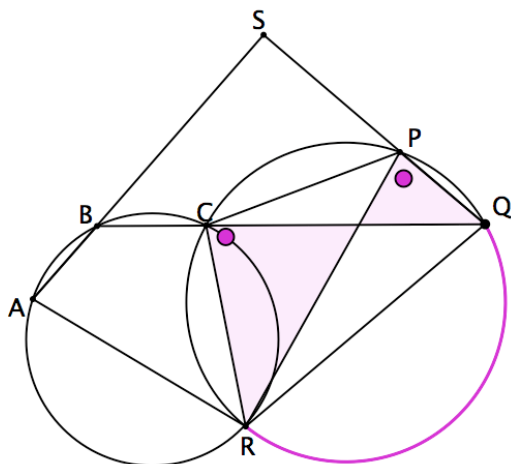
$$= \frac{25\pi}{64} \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_0^2$$

$$= \frac{25\pi}{64} \left[16(2) - \frac{8(2)^3}{3} + \frac{(2)^5}{5} - 0 \right]$$

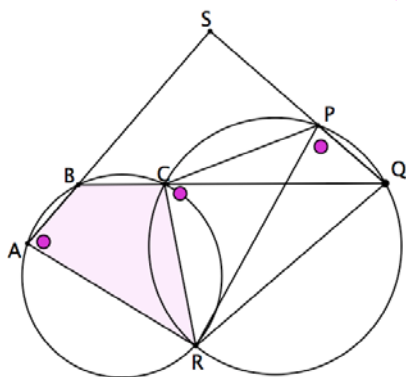
$$\therefore V = \frac{20\pi}{3} \text{ unit}^3$$

(e)

(i)



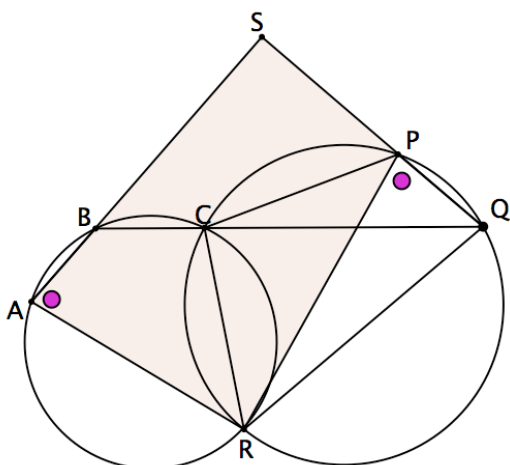
In circle $CPQR$, $\angle QPR = \angle QCR$ (angles standing on the same arc RQ subtend equal angles at the circumference)



In circle $ABCD$, $\angle BAR = \angle QCR$ (exterior angle of a cyclic quadrilateral equals the interior opposite angle)

$\therefore \angle BAR = \angle QPR$

(ii)



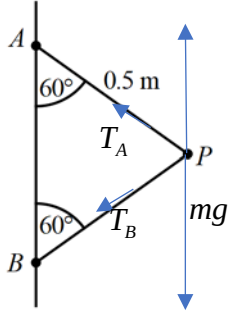
From (i) $\angle BAR = \angle QPR$

$\therefore ARPS$ is a cyclic quadrilateral (exterior angle of a cyclic quadrilateral equals the interior opposite angle)

Question 15

(a)

(i)



$$\sin 60^\circ = \frac{r}{0.5}$$

$$r = \frac{1}{2} \times \frac{\sqrt{3}}{2}$$

$$r = \frac{\sqrt{3}}{4} \text{ m}$$

Vertically:

$$T_A \cos 60^\circ - T_B \cos 60^\circ - mg = 0$$

$$T_A \times \frac{1}{2} - T_B \times \frac{1}{2} = mg$$

$$\frac{1}{2}(T_A - T_B) = mg$$

$$T_A - T_B = mg \quad \dots(1)$$

(ii)

Horizontally:

$$T_A \sin 60^\circ + T_B \sin 60^\circ = mr\omega^2$$

$$\frac{\sqrt{3}}{2}(T_A + T_B) = m \times \frac{\sqrt{3}}{4} \times (2\sqrt{3}g)^2$$

$$\frac{\sqrt{3}}{2}(T_A + T_B) = \frac{\sqrt{3}m}{4} \times 12g$$

$$\frac{1}{2}(T_A + T_B) = \frac{m}{4} \times 12g$$

$$T_A + T_B = 6gm \quad \dots(2)$$

(1)+(2):

$$2T_A = 8gm$$

$$T_A = 4gm \text{ N}$$

$$T_B = 2gm \text{ N}$$

(b)

(i)

$$F = m\ddot{x}$$

$$mg - R = m\ddot{x}$$

$$R = m(g - \ddot{x})$$

$$= m \left(g - \frac{d}{dx} \left(kg \left(1 - e^{-\frac{x}{k}} \right) \right) \right)$$

$$= mg \left(1 - k \times \frac{e^{-\frac{x}{k}}}{k} \right)$$

$$= mg \left(1 - e^{-\frac{x}{k}} \right)$$

$$= \frac{m}{2k} \times 2kg \left(1 - e^{-\frac{x}{k}} \right)$$

$$= \frac{mv^2}{2k}$$

OR

$$F = m\ddot{x}$$

$$mg - R = m\ddot{x}$$

$$\frac{mg - R}{m} = \ddot{x}$$

$$g - \frac{R}{m} = \frac{1}{2} \frac{dv^2}{dx}$$

$$= \frac{1}{2} \frac{d}{dx} \left[2kg \left(1 - e^{-\frac{x}{k}} \right) \right]$$

$$= \frac{d}{dx} \left[kg \left(1 - e^{-\frac{x}{k}} \right) \right]$$

$$= kg \left(\frac{1}{k} e^{-\frac{x}{k}} \right)$$

$$= ge^{-\frac{x}{k}}$$

$$\text{Since } v^2 = 2kg \left(1 - e^{-\frac{x}{k}} \right)$$

$$\frac{v^2}{2kg} = 1 - e^{-\frac{x}{k}}$$

$$e^{-\frac{x}{k}} = 1 - \frac{v^2}{2kg}$$

$$g - \frac{R}{m} = g - \frac{v^2 g}{2kg}$$

$$-\frac{R}{m} = -\frac{v^2}{2k}$$

$$\therefore R = \frac{mv^2}{2k}$$

(ii)

$$m\ddot{x} = -mg - \frac{mv^2}{2k}$$

$$\ddot{x} = -\left(g + \frac{v^2}{2k}\right)$$

$$v \frac{dv}{dx} = -\left(\frac{2kg + v^2}{2k}\right)$$

$$\frac{dv}{dx} = -\left(\frac{2kg + v^2}{2kv}\right)$$

$$\frac{dx}{dv} = -\left(\frac{2kv}{2kg + v^2}\right)$$

$$\begin{aligned} x &= -k \int \frac{2v}{2kg + v^2} dv \\ &= -k \ln(2kg + v^2) + C \end{aligned}$$

Sub $x = 0, v = \sqrt{2kg}$

$$0 = -k \ln(2kg + 2kg) + C$$

$$C = k \ln 4kg$$

$$x = k \ln 4kg - k \ln(2kg + v^2)$$

Let $v = 0$

$$x = k \ln 4kg - k \ln 2kg$$

$$= k \ln\left(\frac{4kg}{2kg}\right)$$

$$= k \ln 2$$

(iii)

$$\ddot{x} = -\left(g + \frac{v^2}{2k}\right)$$

$$\frac{dv}{dt} = -\left(\frac{2kg + v^2}{2k}\right)$$

$$\frac{dt}{dv} = -2k \left(\frac{1}{\left(\sqrt{2kg}\right)^2 + v^2} \right)$$

$$t = \frac{-2k}{\sqrt{2kg}} \tan^{-1} \left(\frac{v}{\sqrt{2kg}} \right) + D$$

When $t = 0$, $v = \sqrt{2kg}$

$$0 = -\frac{2k}{\sqrt{2kg}} \tan^{-1} \left(\frac{\sqrt{2kg}}{\sqrt{2kg}} \right) + C$$

$$C = \frac{2k}{\sqrt{2kg}} \tan^{-1} 1$$

$$C = \frac{2k}{\sqrt{2kg}} \times \frac{\pi}{4}$$

$$= \frac{k\pi}{2\sqrt{2kg}}$$

$$t = -\frac{2k}{\sqrt{2kg}} \tan^{-1} \left(\frac{v}{\sqrt{2kg}} \right) + \frac{\pi k}{2\sqrt{2kg}}$$

Let $v = 0$

$$t = \frac{\pi k}{2\sqrt{2kg}} \text{ seconds.}$$

(c)	
(i) $4x^2 + 3y^2 = 12 \dots(1)$ $y = mx + 4 \dots(2)$ Sub (2) into (1): $4x^2 + 3[m^2x^2 + 8mx + 16] = 12$ $4x^2 + 3m^2x^2 + 24mx + 48 = 12$ $x^2(4 + 3m^2) + 24mx + 36 = 0$ For tangent $\Delta = 0$ $(24)^2 - 4(3m^2 + 4) \times 36 = 0$ $576m^2 - 432m^2 - 576 = 0$ $144m^2 = 576$ $m^2 = 4$ $m = 2 (m > 0)$	
(ii) Sub $m = 2$ into equation above $x^2(3 \times 4 + 4) + 48x + 36 = 0$ $16x^2 + 48x + 36 = 0$ $(2x + 3)^2 = 0$ $x = \frac{-3}{2}$ When $x = \frac{-3}{2}$, $y = 2\left(\frac{-3}{2}\right) + 4$ $= -3 + 4$ $= 1$ $P\left(\frac{-3}{2}, 1\right)$	
(iii) Gradient of tangent is 2, gradient of normal is $-\frac{1}{2}$ Equation of normal is $y - 1 = \frac{-1}{2}\left(x + \frac{3}{2}\right)$ $2y - 2 = -x - \frac{3}{2}$ $4y - 4 = -2x - 3$ $4y + 2x - 1 = 0$ $B(0, 4)$ B	

A: Let $x = 0$ in normal

$$4y - 1 = 0$$

$$y = \frac{1}{4}$$

$$A\left(0, \frac{1}{4}\right)$$

$$A = \frac{1}{2} \left(4 - \frac{1}{4}\right) \times \frac{3}{2}$$

$$= \frac{45}{16} \text{ units}^2$$

Question 16

(a)

(i)

The coefficients of $z^3 + az + 10 = 0$ are real.

Let the roots be α , $1 - 2i$, $1 + 2i$.

Product of the roots:

$$\alpha(1 - 2i)(1 + 2i) = -10$$

$$5\alpha = -10$$

$$\alpha = -2$$

Sum of roots two at a time:

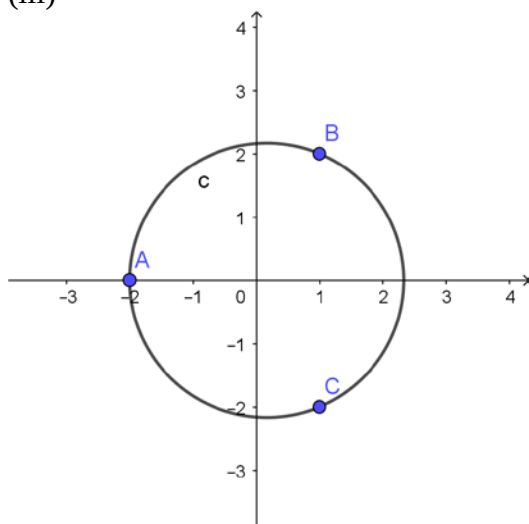
$$\alpha(1 - 2i) + \alpha(1 + 2i) + (1 + 2i)(1 - 2i) = a$$

$$2\alpha + 5 = a$$

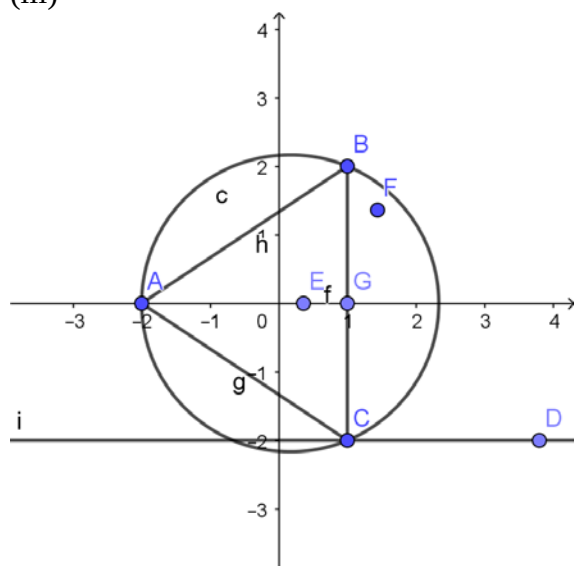
$$a = 1$$

$$z^3 + az + 10 = 0$$

(iii)



(iii)



By symmetry, centre must be on the x-axis.

Construct CD parallel to the x-axis.

$$\arg(-2 - (1 - 2i)) = \arg(-3 + 2i)$$

$$= \pi - \tan^{-1} \frac{2}{3}$$

$$\angle DCA = \pi - \tan^{-1} \frac{2}{3}$$

By co-interior $\angle$'s

$$\angle CAE = \pi - \left(\pi - \tan^{-1} \frac{2}{3} \right)$$

$$= \tan^{-1} \frac{2}{3}$$

$$\angle CEG = 2 \tan^{-1} \frac{2}{3} \left(\begin{array}{l} \text{angle at the centre is twice the} \\ \text{angle at the circumference} \end{array} \right)$$

In $\triangle CEG$

$$\sin \theta = \frac{2}{r}$$

$$r = \frac{2}{\sin \left(2 \tan^{-1} \frac{2}{3} \right)}$$

$$= \frac{13}{6}$$

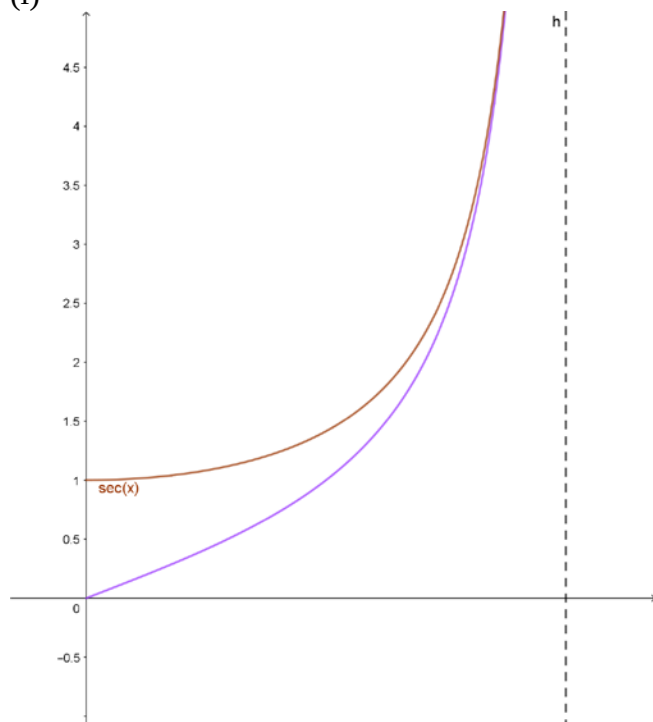
The centre is $\left(\frac{1}{6}, 0 \right)$

$$\text{Equation of locus is } \left| z - \frac{1}{6} \right| = \frac{13}{6}$$

(b)	
(i) $(\sqrt{a} - \sqrt{b})^2 \geq 0$ $(\sqrt{a})^2 - 2\sqrt{a}\sqrt{b} + (\sqrt{b})^2 \geq 0$ $a + b \geq 2\sqrt{ab}$ $\frac{a+b}{2} \geq \sqrt{ab}$	
(ii) $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$ From what is given $\sqrt[n]{n \times (n-1) \times (n-2) \times \dots \times 2 \times 1} \leq \frac{n + (n-1) + (n-2) + \dots + 2 + 1}{n}$ $\sqrt[n]{n!} \leq \frac{1}{n} \times \frac{n}{2} (n+1)$ $\sqrt[n]{n!} \leq \frac{n+1}{2}$ $n! \leq \left(\frac{n+1}{2} \right)^n$	

(c)

(i)



(ii)

$$LHS = \sec \theta - \tan \theta$$

$$= \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin \theta}{\cos \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos \theta (1 + \sin \theta)}$$

$$= \frac{\cos^2 \theta}{\cos \theta (1 + \sin \theta)}$$

$$= \frac{\cos \theta}{1 + \sin \theta}$$

$$= \frac{1}{\sec \theta (1 + \sin \theta)}$$

$$= \frac{1}{\sec \theta + \frac{\sin \theta}{\cos \theta}}$$

$$= \frac{1}{\sec \theta + \tan \theta}$$

OR

$$LHS = \sec \theta - \tan \theta$$

$$= \sec \theta - \tan \theta \times \frac{\sec \theta + \tan \theta}{\sec \theta + \tan \theta}$$

$$= \frac{\sec^2 \theta - \tan^2 \theta}{\sec \theta + \tan \theta}$$

$$= \frac{1}{\sec \theta + \tan \theta}$$

(iii)

Given that $\sec \theta > \tan \theta$ for $0 \leq \theta < \frac{\pi}{2}$

$$\sec \theta > \tan \theta$$

$$\sec \theta - \tan \theta > 0$$

$$\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

From graph $\sec \theta + \tan \theta \geq 1$

$$1 \geq \frac{1}{\sec \theta + \tan \theta}$$

$$\therefore \sec \theta - \tan \theta \leq 1$$

$$0 < \sec \theta - \tan \theta \leq 1$$

OR

As graphically shown, $y_2 - y_1 = \sec \theta - \tan \theta$

and $\sec \theta - \tan \theta > 0$ for $0 \leq \theta < \frac{\pi}{2}$

At $\theta = 0$, $\sec \theta - \tan \theta = 1$ i.e. the distance is maximum

As $\theta \rightarrow \frac{\pi}{2}$, $(\sec \theta - \tan \theta) \rightarrow 0$

Hence $0 < \sec \theta - \tan \theta \leq 1$.

(iv)

$$\sec \theta - \tan \theta = \frac{1}{2} \dots (1)$$

$$\frac{1}{\sec \theta + \tan \theta} = \frac{1}{2}$$

$$\sec \theta + \tan \theta = 2 \dots (2)$$

(1) + (2)

$$2 \sec \theta = \frac{5}{2}$$

$$\sec \theta = \frac{5}{4}$$

$$\cos \theta = \frac{4}{5}$$

$$\theta = \cos^{-1} \frac{4}{5}$$

$$\theta = 2k\pi \pm \cos^{-1} \left(\frac{4}{5} \right)$$